

SWANet: A Sliding Window Affinity Network With Dual-Branch Structure for Detection of Automotive Chip Packaging Carriers

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Abstract—In automotive chip packaging carrier (ACPC) defect detection, distinguishing defects is particularly challenging due to the high similarity between missing regions and background textures. Additionally, the scarcity of defect samples further exacerbates this issue. To address these problems, a defect detection network based on sliding window affinity (SWA) with a dual-branch structure has been proposed. The defect sample pool is effectively expanded through a region transformation augmentation technique, which helps alleviate class imbalance issues. Affinity maps are calculated via local affine transformations to enhance internal pixel relationships, and heterogeneous feature parsing (HFP) is employed to emphasize subtle proactive layer features. This method achieves a significant improvement in detection accuracy, reaching 74.38% mAP on the ACPC dataset, and demonstrates strong robustness under varying noise conditions. The results indicate that the proposed method meets the requirements of real-world production environments and has the potential for application in other industrial scenarios.

Index Terms—Automotive chip packaging carriers (ACPCs), cross-region transformation, detection, dual-branch, sliding window affinity (SWA).

I. INTRODUCTION

WITH the scarcity of oil and the increasing emphasis on environmental protection in various countries, new energy vehicles and electric vehicles have been developing rapidly in recent years. As the core of controlling most of the application scenarios of electric vehicles, chips are also the most important factor in determining the safety, comfort, and service life of the entire vehicle. Automotive chip packaging carriers (ACPCs), as the carrier for placing IC chips, are the key to ensuring the normal operation of electric vehicle chips. They provide a stable and reliable operating environment for electric vehicle chips, shielding them from physical shocks, temperature fluctuations, humidity, and other environmental factors, as well as electromagnetic interference. Additionally,

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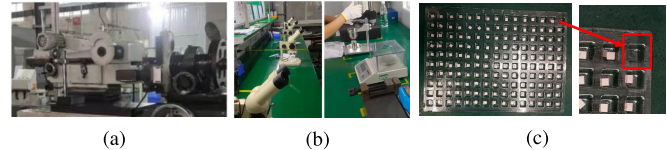


Fig. 1. ACPCs production line. Manually detecting missing positions in assembly boxes. (a) Production equipment, (b) manual assembly, and (c) ACPCs missing assembly detection.

they prevent the integrated circuits from overheating, thereby reducing the likelihood of failures and the need for frequent replacements.

Fig. 1 shows the actual production line of ACPCs. The ACPCs are first cut by the relevant production equipment and then loaded into the test cassettes by the assembly equipment. During the assembly process, ACPCs may be missed due to aging of the machine or other factors, which affect the production efficiency. Detecting the missing assembly of ACPCs is of great significance to reduce production costs and improve production efficiency. At present, the missing detection of ACPCs is mainly performed manually by skilled workers. However, this detection method is labor-intensive, inefficient, and inaccurate. It is crucial to research machine vision-based detection of missing assembly of ACPCs. In general, machine vision-based missing assembly detection can be regarded as a special object detection task.

Currently, deep learning [1], [2] methods have been widely applied in industrial product inspection. While dedicated research on the automated detection of missing components in ACPCs remains relatively limited, related defect detection studies offer valuable insights. Similar scenarios can be referenced in surface defect detection based on machine vision, such as battery defect detection [3], [4], [5], steel strip defect detection [6], [7], [8], and electric two-wheeled vehicle defect detection [9]. For example, Badmos et al. [3] learned an accurate representation of microstructural images by the CNN to accurately distinguish between defective and nondefective cases. Zhao et al. [6] improved the traditional fast R-CNN by using a multiscale algorithm, replacing the traditional convolution with a deformable convolution to improve the accuracy of the detection of defects in steel belts. In another study, a convolutional neural network (CNN) steered with wavelet synchro squeezing transform (WSST)-based scalograms is proposed for fault identification in an

electric two-wheel [9]. Zhang et al. [10] proposed an insulator defect detection framework based on image reconstruction, which achieved high-precision detection of various insulator defects by collecting images of catenary insulators captured by high-speed rail inspection equipment and constructing the catenary insulator defect (CID) dataset. However, given that catenary insulators are large detection objects, this method is not suitable for small object detection tasks like those involving ACPCs.

Although these methods have achieved promising results in industrial product inspection, several challenges are encountered when directly applied to ACPCs missing component detection.

- 1) There is a scarcity of samples in the ACPC's missing component regions, resulting in the issue of sample imbalance. Several methods have been employed to enhance data and increase sample diversity. CutPaste [11], for instance, offers a way to generate defect samples by randomly copying and pasting regions of the original samples to augment the data. However, due to the ordered arrangement of ACPCs, directly applying this method would disrupt the actual distribution of samples, resulting in a significant difference from real test samples, thereby limiting generalization.
- 2) The pixel proportion of missing regions in ACPCs is small, making it difficult to effectively extract discriminative features. Some multimodal approaches address similar small-object issues through multiscale fusion. For example, Zhu et al. [12] proposed an efficient multiscale perception enhancement network (ADDet) for detecting aluminum defects, considering the inherent characteristics of aluminum surface defects, which exhibit diverse morphologies and multiscale variations. Zheng et al. [13] introduced a novel efficient conflict filtering network (ECF-Net) specifically for detecting small defects. However, these methods are challenging to apply directly to ACPCs, where the difference in texture and color between the detection object and the background is minimal, making such tasks more complex.
- 3) The proactive layer and supporting layer of ACPCs are highly similar, making direct differentiation challenging. Existing Transformers are primarily designed for global information modeling, while CNNs excel at capturing local information. ACPCs, built upon this foundation, must distinguish between the proactive layer and the supporting layer under conditions where their colors closely resemble those of the background, resulting in minimal pixel variance across regions. The mere use of CNNs or Transformers struggles to yield robust feature discriminability; additionally, the pixel features in the missing areas are not pronounced, often overlooked in the feature extraction by existing CNNs or Transformers, complicating precise localization.

To address these challenges, a sliding window affinity network (SWANet) with a dual-branch structure is proposed in this article for the detection of ACPCs. The network

incorporates a cross-region transformation algorithm (CRTA), a sliding window affinity (SWA) module, a dual-branch structure, and a heterogeneous feature parsing (HFP) module. The CRTA is employed to augment sample data and mitigate overfitting by randomly transforming missing regions and background segments. The SWA module enhances the detection of missing component regions by calculating local feature affinity through a sliding window mechanism. The dual-branch structure is designed to extract the proactive layer and supporting layer features separately, thereby addressing the issue of high similarity between the proactive layer and the supporting layer in ACPC images. The HFP module parses the features extracted by the dual branches and strengthens the correlation between the proactive layer and the supporting layer using a cross-attention (CA) mechanism, facilitating precise localization of missing regions. To further improve detection accuracy, multiscale feature decoding is utilized to fuse deep and shallow features for the accurate detection of small objects.

In summary, the main contributions of this article are as follows.

- 1) To tackle the issues of data imbalance and the scarcity of defective samples, the CRTA has been introduced, utilizing regional random transformations and positional exchanges to augment the dataset and enhance the network's generalization capability. This data augmentation technique effectively mitigates class imbalance and improves the network's robustness under varying noise conditions.
- 2) The SWA has been proposed, which enhances pixel interconnections in object regions through pixel affinity calculations in local areas. This mechanism not only effectively captures both local and global contextual information but also improves the detection accuracy of small objects in ACPCs, demonstrating exceptional performance in scenarios where pixel features exhibit high similarity.
- 3) An innovative dual-branch structure has been designed to separately extract proactive layer and supporting layer features, addressing the high similarity between these elements in ACPCs. Through the parsing conducted by the HFP module, this dual-branch structure effectively amplifies the differences between the proactive layer and the supporting layer, thereby enhancing the network's ability to discern missing components and subtle variations, particularly in complex industrial environments.

The rest of the article is organized as follows. Related work is presented in Section II. SWANet is discussed in Section III. The constructed dataset and related experiments are presented in Section IV. Section V presents the conclusions of this study.

II. RELATED WORK

This study is related to the application of computer vision techniques for missing assembly detection of ACPCs. In turn, missing assembly detection serves as an important category in the defect detection of ACPCs. Currently, there are fewer missing assembly detections for ACPCs. Similar work can

be referred to as industrial surface defect detection, which is categorized into traditional methods, CNN-based methods, and Transformer-based methods.

A. Traditional Methods for Defect Detection

Traditional methods have continued to play an important role in industrial surface defect detection for a long time, mainly through manually extracted features for defect detection. Currently, traditional defect detection methods are categorized into three main types: 1) statistical-based defect detection methods [17], [18], [19], [20]; 2) structure-based defect detection methods [21], [22], [23], [24]; and 3) model-based defect detection methods [25], [26], [27], [28]. The first method consists of learning models from data to differentiate between normal data objects and anomalies for defect detection. Yuan et al. [29] proposed an improved Otsu method for detecting defects on product surfaces by weighted object variance (WOV). However, the method depends heavily on the key point detection algorithm or similarity measurement strategy. The second method focuses on the spatial location of pixels. Tolba and Raafat [30] proposed the multiscale structural similarity index (MS-SSIM), which shows the strong discriminative ability of MS-SSIM between normal and abnormal surfaces. The computational complexity of the third method is strongly affected by the estimation of the stochastic model parameters. Wang et al. [17] applied the entity sparsity pursuit (ESP) method to identify surface defects based on the fact that surface image textures usually form a low-rank structure.

B. Deep Learning Methods for Defect Detection

In the past few years, the emergence of deep learning algorithms has brought deep transformations to machine learning algorithms, distinguishing them from traditional methods such as eddy current inspection [31] and ultrasonic inspection [32] by their convenience and speed. For example, for the surface quality inspection problem of LED chips, Shu et al. [33] proposed a parallel DCNN model and built a fully automated inspection system operating with high detection accuracy (99%) to solve the cover glass defect detection problem. Lu et al. [34] presented a multiscale feature-enhanced fusion (MFEF) for surface defects detection and a named MRD-Net reverse attention (RA) network for real-time and end-to-end defect segmentation. Liang et al. [35] used deep dynamic convolution as the basic structure of the network to detect surface defects of nickel-plated perforated steel strips using adaptive feature extraction. Li et al. [36] suggested an automated vision and line-laser feeding system combining industrial object localization and classification methods to detect defects in the assembly of crankshaft shingle covers.

Recently, due to the remarkable performance of Transformers in computer vision, a multitude of studies have been devoted to introducing Transformers into defect detection tasks. In contrast to CNNs, where the inherent localization of convolution prevents them from explicitly modeling long-range interactions, Transformer-based networks can correlate each element of the data with each other through their built-in global self-attention mechanism. Zhu et al. [37] proposed

a novel network architecture called LSwin Transformer to construct a convolutional embedding module and an attention patch merging module to improve the retention of image information. Zhang et al. [38] proposed a two-stage “promote-suppress” Transformer (PST) framework, which employs wavelet features to direct the network to focus on the detailed features in the image and accurately localize and classify the defective regions in the image. Zhou et al. [39] effectively classified and sensed defects of different shapes and sizes by fusing two modules, ETDNet, a Transformer-based detection network, and a channel-modulated feature pyramid network (CM-FPN). Gao et al. [14] constructed a novel Transformer detection framework for surface defect detection of automotive bearings and so on, which outperforms CNN detection. Luo et al. [15] addressed the problem of semantic gaps between multiscale features by importing the CA Transformer (CAT) into the encoder-decoder network (EDNet). Yu et al. [40] presented a dynamic Transformer network for surface defect detection, which utilizes a self-attention-based Transformer to accurately capture long-range semantic information in the routing space, enabling accurate online detection of multiscale surface defects.

C. Anomaly Detection Methods

Additionally, anomaly detection methods have become an important research direction in industrial scenarios. In recent years, both self-supervised and unsupervised learning approaches have made significant progress in this field. The DRAEM method proposed by Zavrtanik et al. [41] utilizes an anomaly synthesis strategy within a self-supervised learning framework to enhance surface anomaly detection performance. The anomaly detection method based on a self-supervised predictive convolutional attentive block (SCAB) proposed by Ristea et al. [42] has also shown excellent performance, combining prediction with self-supervised learning to improve the detection accuracy of industrial defects. The unsupervised anomaly detection method based on a masked convolutional Transformer block proposed by Madan et al. [43] further demonstrates how convolution and Transformer can be combined under unsupervised conditions to enhance the model’s ability to perceive defect regions. Building on this, Roth et al. explored how to improve anomaly detection performance in industrial settings through a global recall mechanism, proposing PatchCore, which effectively balances memory-efficient feature sampling with high recall performance in industrial anomaly detection [44]. Additionally, Liu et al. [45] proposed the appearance-motion prototype network (AMP-net), which leverages external memory mechanisms to capture appearance and motion features, excelling in anomaly detection for surveillance videos. This approach balances the effective representation of normal events and anomaly detection, achieving state-of-the-art (SOTA) performance [45]. Liu et al. [46] also developed the memory-enhanced spatial-temporal encoding (MSTE) framework, specifically designed for industrial scenarios, addressing the challenges of scarce industrial video data and the complexity of anomaly events, improving the accuracy and spatial-temporal localization of anomaly

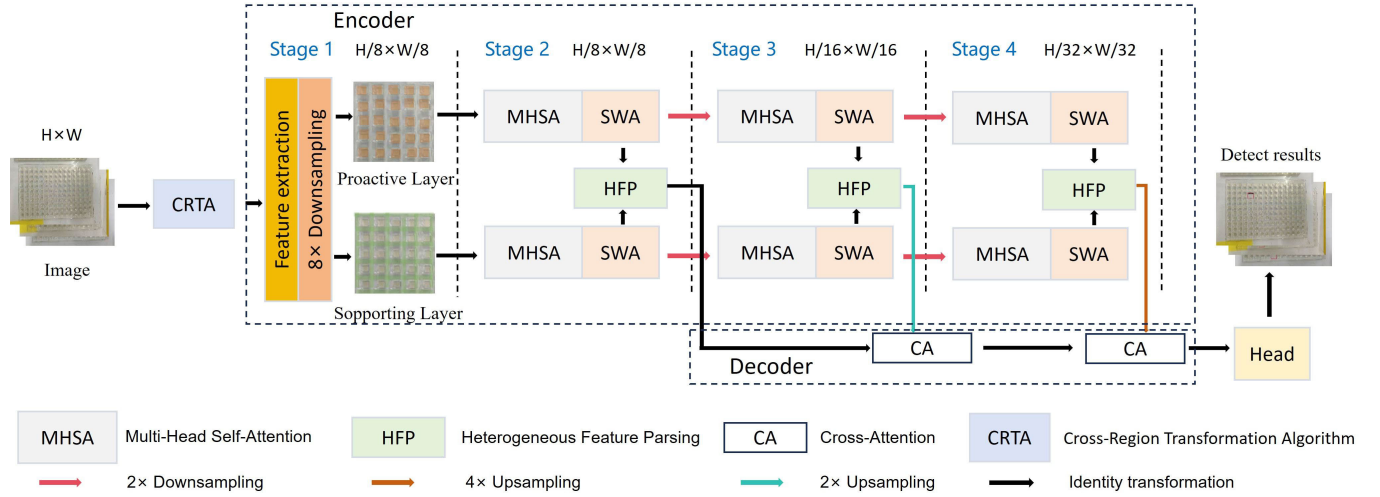


Fig. 2. Pipelines of the detection network. Stage 1 is used for feature extraction; Stages 2–4 are for multiscale feature learning, including MHSA, SWA, and HFP.

detection. Gong et al. [47] proposed the memory-augmented autoencoder (MemAE), which enhances the ability to memorize normal patterns by incorporating a memory module into the autoencoder architecture, thereby improving anomaly detection performance. Wang et al. [48] introduced a dual-branch learning framework with prior information for surface anomaly detection, which effectively mitigates the overkill problem, a common issue in industrial defect detection. Liu et al. [49] developed SimpleNet, a lightweight yet effective model designed to enhance the compatibility between defect detection models and industrial imaging, thereby improving detection accuracy. Li et al. [50] proposed a novel approach for few-shot anomaly detection by leveraging prompt learning based solely on normal samples, enabling more efficient learning in scenarios with limited anomaly data.

III. PROPOSED METHOD

A. Structure of the Detection Model

In this article, a SWANet with a dual-branch structure is proposed for the detection of missing components in ACPCs. Initially, the data undergoes augmentation via the CRTA before being fed into the network. As illustrated in Fig. 2, the network employs an encoder–decoder architecture. Unlike traditional object detection networks, the proposed encoder utilizes a dual-branch structure to separately extract and process distinctive features of the proactive layer and supporting layer, significantly enhancing the network’s capability to perceive subtle differences between these two elements, thereby addressing the issue of high similarity in ACPC images.

In the dual-branch structure, the proactive layer and the supporting layer, respectively, take on the roles of capturing core information and providing auxiliary support. This division enables the network to more effectively distinguish key features and background details in complex scenarios, thereby enhancing its ability to detect missing components, especially in ACPC images with high-similarity features. Each branch comprises four feature extraction and processing stages. Specifically, the input image is downsampled to

one-eighth of its original size in the first stage. Unlike the first stage, the second to fourth stages perform a $2\times$ downsampling between adjacent stages. Moreover, to address variations in angle and scale present in the captured images, the network incorporates a multiscale structure. Three MHSA-SWA structures are designed to learn the multilevel features of ACPCs. In the MHSA-SWA structure, the global modeling capability of the Transformer is utilized to extract discriminative global features for the proactive layer $F_i (i = 1, 2, 3)$ and supporting layer $B_i (i = 1, 2, 3)$. These global features are then input into the corresponding SWA modules of each stage, where pixel affinity relationships are established through a sliding window mechanism.

The features of the proactive layer and the supporting layer are first obtained by mapping to high-dimensional embeddings through a weight-sharing CNN. Subsequently, two MHSA branches with nonshared weights are constructed to extract the proactive layer and supporting layer features separately. This overall CNN–Transformer structure effectively captures detailed information about the object while establishing comprehensive global connections.

To parse the heterogeneous features within the dual branches, the HFP module is constructed to accurately differentiate between the proactive layer and the supporting layer, allowing the network to focus on detecting assembly differences of the missing components. During the decoding phase, the multiscale HFP features are progressively decoded through CA, facilitating the gradual fusion of features across different levels. These discriminative features are directly input into the detection head for identifying missing regions within the ACPC assembly.

B. Cross-Region Transformation Algorithm

Due to the small number of missing samples in the collected ACPCs dataset, there is a problem of imbalance between positive and negative samples, which will cause over-fitting. Therefore, to avoid the above problems, it is necessary to enhance the collected image data to achieve the purpose of

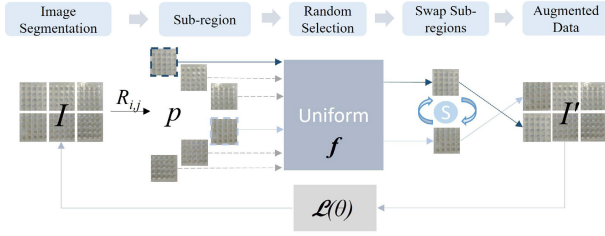


Fig. 3. Pipelines of CRTA. The input image is divided into six equally sized subregions. The random function f and probability matrix p are used for exchanging subregions. Optimize parameters θ through loss function L .

expanding the data. Compared with only using data augmentation methods such as color transformation, this article can change the pixel distribution of the whole image by using the above local region transformation method, so that the network pays more attention to the difference between the missing area and the normal area, thereby improving the generalization ability of the model.

Before initiating the data augmentation step, it is essential to extract the object region from the collected images to facilitate subsequent subregion swapping. As illustrated in the figure, the input image I is set with dimensions $H \times W$ and divided into $m \times n$ areas with each area sized $((H/m), (W/n))$, where $m = 2$ and $n = 3$. The specific process is shown in Fig. 3.

Each subregion $R_{i,j}$ is defined by its row i and column j indices and can be specifically expressed as

$$R_{i,j} = I \left[i \cdot \left(\frac{H}{m} \right) : (i+1) \cdot \left(\frac{H}{m} \right) \right. \\ \left. j \cdot \left(\frac{W}{n} \right) : (j+1) \cdot \left(\frac{W}{n} \right) \right]. \quad (1)$$

Following this, two different subregions (i_1, j_1) and (i_2, j_2) are randomly selected, ensuring they are distinct. This selection process can be mathematically expressed as

$$(i_1, j_1) \sim \text{Uniform}(1, m) \times \text{Uniform}(1, n) \\ (i_2, j_2) \sim \text{Uniform}(1, m) \times \text{Uniform}(1, n) \\ (i_1, j_1) \neq (i_2, j_2). \quad (2)$$

A swap operation S is defined, where the selected areas R_{i_1, j_1} and R_{i_2, j_2} are exchanged. The image after swapping, I' , is represented as

$$I'(R_{i_1, j_1}) = R_{i_2, j_2} \\ I'(R_{i_2, j_2}) = R_{i_1, j_1} \quad (3)$$

with other areas remaining unchanged.

The swap position selection function f is defined as $(i_1, j_1), (i_2, j_2) = f(p; \theta)$, where p is a probability matrix representing the likelihood of each area being selected, learned through training network parameters θ .

Finally, a loss function L is defined based on the L_2 loss, which minimizes the pixel-wise difference between the transformed image I' and the target image I . It is formulated as: θ

$$L(\theta) = \|I' - I\|^2 \quad (4)$$

Algorithm 1 Random Region Exchange Module

- 1: **Input:** Image I of size $H \times W$, divided into $m \times n$ regions, network parameters θ
- 2: **Output:** Augmented image I'
- 3: Divide I into regions $R_{i,j}$ where $i \in \{1, \dots, m\}$ and $j \in \{1, \dots, n\}$:

$$R_{i,j} = I \left[i \cdot \frac{H}{m} : (i+1) \cdot \frac{H}{m}, j \cdot \frac{W}{n} : (j+1) \cdot \frac{W}{n} \right]$$

- 4: Generate probability matrix $\mathbf{p}$ using the network $f(\mathbf{p}; \theta)$
- 5: Randomly select two regions (i_1, j_1) and (i_2, j_2) based on $\mathbf{p}$ such that $(i_1, j_1) \neq (i_2, j_2)$
- 6: Perform region exchange:

$$I'(R_{i_1, j_1}) = R_{i_2, j_2} \\ I'(R_{i_2, j_2}) = R_{i_1, j_1}$$

- 7: Keep other regions unchanged
- 8: Compute the loss $\mathcal{L}(\theta)$ using I' and I
- 9: Update θ to minimize the loss:

$$\theta \leftarrow \theta - \eta \cdot \nabla_{\theta} \mathcal{L}(\theta)$$

- 10: **Return:** Augmented image I'

where I is the object image, guiding the network to learn how to enhance data through region swapping and improve the model's generalization capability.

To avoid the disruption of data distribution consistency caused by random region transformations, we supervise the similarity between regions before and after transformation based on the aforementioned mathematical model. The specific process is detailed in Algorithm 1, where the transformation parameters θ are set to be learned through the network. This adaptive transformation parameter learning approach prevents excessive region transformations that could lead to abnormal data distribution, ensuring the expansion of samples while maximizing the similarity between the original and augmented samples.

C. Sliding Window Affinity

To make the network learn the location of the missing area in ACPCs more quickly, we use a new method to calculate the similarity between the object pixel area and the nonobject pixel area, that is, the SWA method. The calculation result of the similarity of different regions is affinity. The sliding window size of the sliding affinity module is set to be 5×5 , and the sliding step size is 1. First, the local window of 5×5 in the upper left corner of the feature map is selected. In the feature map part of the selected local window, linear mapping is performed to obtain the Q and the K , and the local affinity is obtained by multiplying Q and K . Finally, the local affinity is mapped to the original position of the feature map. The calculation process is shown in Fig. 4.

The feature map of 28×28 sizes obtained after feature fusion is connected to the sliding affinity module. The window slides on the original feature map according to the step size. The feature map selected by the sliding window continues

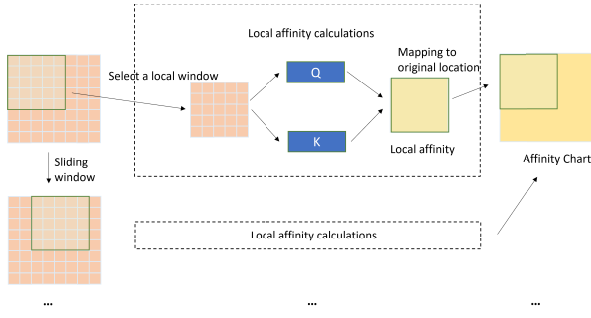


Fig. 4. Pipelines of SWA. A 5×5 local window in the feature map is selected and linearly mapped to obtain Q and K , from which the local affinity is computed and mapped back to the original position. The sliding window traverses the feature map with a stride of 1, repeatedly computing and mapping the local affinity. For overlapping regions, affinities are multiplied, ultimately generating a complete affinity map.

to calculate the local affinity and is also mapped to the corresponding position of the feature map. If there is an overlapping part, the affinity is multiplied. Then, repeat the steps of sliding, calculating affinity, and mapping to the feature map until the sliding is completed, and the affinity map of the entire feature map is obtained.

Let f be the original feature map, w_f and h_f be the width and height of f , respectively, and the feature map of each local window divided by f is

$$f_{ij} = f[5i : 5i + 5/5, 5j : 5j + 5/5] \quad (5)$$

$$i = 0, 1, \dots, w_f - 5, \quad j = 0, 1, \dots, h_f - 5. \quad (6)$$

The local affinity diagram calculated by SWA is

$$\text{affi}(f_{ij}) = \text{softmax}\left(Q \times K^T / \sqrt{d_k}\right). \quad (7)$$

Among them, Q , K , and V are obtained by the linear transformation of the characteristic map f

$$Q = w_q f_{ij} \quad (8)$$

$$K = w_k f_{ij} \quad (9)$$

$$V = w_v f_{ij}. \quad (10)$$

Finally, the affinity map of the characteristic map f is obtained as

$$\text{Affi}(f) = \sum_{i,j} \text{affi}(f_{ij}) \times \alpha_{ij}. \quad (11)$$

Among them, α_{ij} is the weight coefficient, which determines the proportion of local affinity in the affinity map at each position.

D. Heterogeneous Feature Parsing

HFP is intended to address the challenges associated with the high similarity between ACPC assembly regions and their background, a similarity that complicates the detection of missing components. This resemblance makes it difficult for traditional methods to accurately distinguish between the regions. The core design principle of the HFP module lies in processing the heterogeneous proactive layer and supporting layer features separately, followed by the application of an attention mechanism to prioritize the more salient proactive

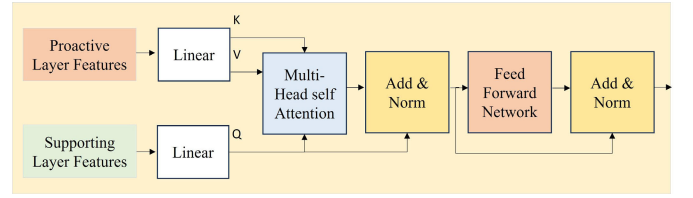


Fig. 5. Pipelines of HFP. The proactive layer and the supporting layer features undergo linear transformations, with the supporting layer generating queries (Q) and the proactive layer generating keys (K) and values (V). A multihead self-attention mechanism captures the feature relationships, focusing on subtle differential regions. The output features are processed through a feedforward network and normalization layers.

layer features. This approach allows the network to effectively learn distinguishing characteristics for each region, thereby improving the model's ability to detect subtle differences in the assembly areas.

As illustrated in Fig. 5, the features are first processed through their respective linear layers to ensure that each set undergoes appropriate transformation before the attention computation. A multihead CA mechanism is then employed, where the supporting layer features are used to generate queries (Q), while the proactive layer features produce keys (K) and values (V). This configuration enables the network to learn the relationships between the proactive layer and the supporting layer, allowing it to focus on regions where the differences, though subtle, are crucial. The use of the self-attention mechanism aids the network in capturing long-range dependencies, which is particularly beneficial in identifying the anomalies related to missing components in ACPC assembly. The output features are subsequently processed by a feedforward network and normalization layers to ensure that the parsed features maintain a high level of discriminative power. This design significantly enhances the model's performance in distinguishing between the proactive layer and the supporting layer, especially in environments where the assembly regions closely resemble the background.

E. Loss Function

To jointly optimize both classification and bounding box regression, we combine focal loss and CIoU loss using a weighted sum. The total loss function is defined as

$$\mathcal{L}_{\text{total}} = \lambda_1 \mathcal{L}_{\text{focal}} + \lambda_2 \mathcal{L}_{\text{CIoU}} \quad (12)$$

where $\mathcal{L}_{\text{focal}}$ is the focal loss for classification, $\mathcal{L}_{\text{CIoU}}$ is the CIoU loss for bounding box regression, and λ_1 and λ_2 are the weights that control the tradeoff between the classification and localization loss, respectively.

Focal loss is used to reduce the impact of easily classified negative samples in ACPCs and focus more on hard-to-classify positive samples. It reshapes the standard cross-entropy loss by lowering the weight of correctly classified examples and increasing the weight of misclassified or difficult examples. The definition of focal loss is as follows:

$$\mathcal{L}_{\text{focal}}(p_t) = -\alpha_t (1 - p_t)^\gamma \log(p_t) \quad (13)$$

based on empirical experience, we set $\alpha_t = 0.25$ and $\gamma = 2$.

CIoU loss is used to improve the localization accuracy of bounding boxes. This loss function not only considers the overlap between the predicted and ground-truth boxes of ACPCs but also takes into account their aspect ratios and the distance between their center points. The definition of CIoU loss is as follows:

$$\mathcal{L}_{\text{CIoU}} = 1 - \text{IoU} + \frac{\rho^2(\mathbf{b}, \mathbf{b}^g)}{c^2} + \alpha v. \quad (14)$$

Here, $\mathbf{b}$ and $\mathbf{b}^g$ are the center points of the predicted and ground-truth boxes, ρ is the Euclidean distance between these points, c is the diagonal length of the smallest enclosing box covering both the predicted and ground-truth boxes, v measures the similarity of aspect ratios, and α is a positive tradeoff parameter.

IV. EXPERIMENT

In this section, we show the evaluation results of the proposed method on the ACPC dataset, which validates the effectiveness of our proposed method.

A. Dataset and Environment

This study utilized an ACPC's image acquisition system, with the collection process illustrated in Fig. 6: ACPCs were placed flat on a platform and illuminated by a single LED light source. In this system, a Basler ruL2098-10gc charge-coupled device (CCD) industrial camera was used as the acquisition sensor, and a display screen was used for real-time monitoring of the acquisition process. Images with missing assembly are manually filtered out for the object box labeling. Due to the presence of different production lines for ACPCs, there is no single type of chip for missing assembly detection, posing higher requirements for the missing detection network. Fig. 7 shows two missed and nonmissed chip maps with different backgrounds were selected for gray value detection, respectively, and the results show that the gray values of the missed and nonmissed maps with different backgrounds are very close to each other, which illustrates the characteristics of the dataset in which the differences between the missed and nonmissed regions are small.

Through the ACPC's image acquisition system and data augmentation algorithms, the ACPC dataset was established. The images in the ACPC dataset were uniformly resized to 2560×1920 pixels to ensure consistency in model training and testing. The dataset contains 1228 labeled missing images, with 983 used for training and 245 for testing. In the missing images, the average pixel proportion of the missing area is 1%.

The experiment is carried out on the Linux Ubuntu 20.4 platform. The GPU is NVIDIA Tesla A100, and the deep learning environment is python3.8 + pytorch1.8 + cuda11.4. The learning rate is 0.0001, epoch is 100, and the optimizer is Adam.

B. Evaluation Metrics

In this study, the training loss curves, mean average precision (mAP), frames per second (FPS), and other relevant metrics are used to evaluate the performance of the model.

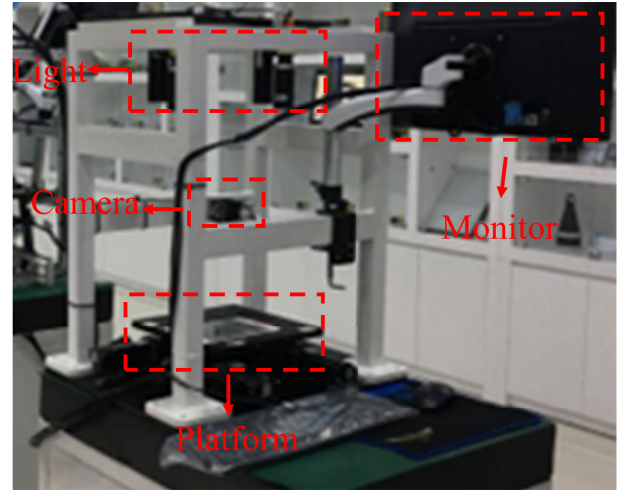


Fig. 6. ACPC image acquisition system using Basler CCD industrial camera, demonstrating the process of flat placement for missing assembly detection.

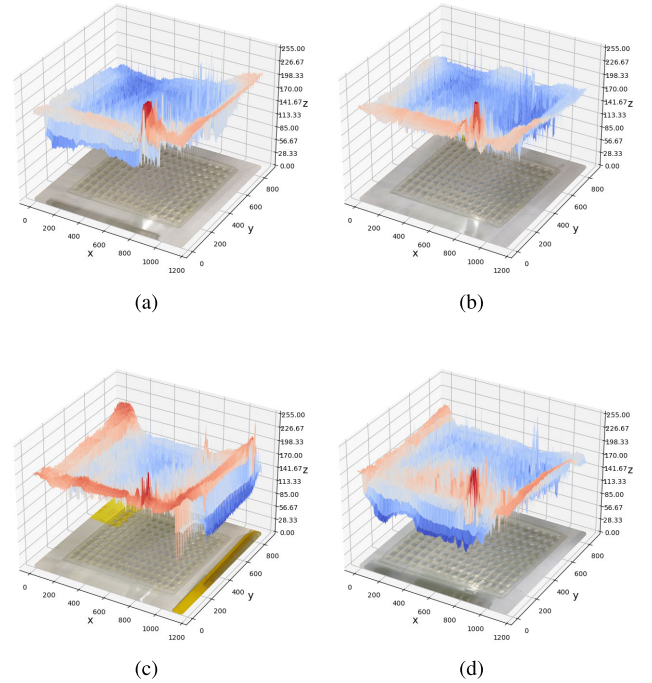


Fig. 7. Comparison of gray values between (a) and (b) missing and (c) and (d) nonmissing areas of ACPCs on different backgrounds.

1) *Training Loss Curves*: The curve represents the variation of the model's performance metrics on the training set with the number of training samples. It is a curve with dataset size as the horizontal axis and model accuracy as the vertical axis. By gradually increasing the dataset size and observing the trend of accuracy change on the training set, the optimization space and overfitting of the model can be known. The performance of the model can be improved by adjusting the model complexity or obtaining more data.

2) *Mean Average Precision*: The average precision (AP) can be obtained in object detection tasks by calculating the area under the $P-R$ curve, reflecting the average accuracy rate for different recall cases. AP50 is one of the commonly used metrics, and it represents the AP value when the intersection over the union (IoU) threshold is 50%. In this article, AP50 is

TABLE I
PERFORMANCE COMPARISON OF VARIOUS DETECTION METHODS FOR ACPC MISSING ASSEMBLY DETECTION BASED ON MAP AND FPS

Method	map50	map75	map50:95	mar50:95	FPS	Params	GFLOPs
Swin Transformer [37]	0.971± 0.021	0.899± 0.055	0.6762± 0.020	0.740± 0.0115	76.1	39.8	268.5
Faster RCNN [51]	0.948± 0.028	0.5256± 0.0571	0.533± 0.0161	0.6054± 0.0066	61.1	42.0	284.5
FCOS [52]	0.7642± 0.1188	0.319± 0.1711	0.3936± 0.0887	0.5556± 0.0738	65.4	30.9	139.4
RetinaNet [53]	0.955± 0.0107	0.874± 0.0157	0.672± 0.0281	0.710± 0.0221	69.5	34.5	151.1
Ours	0.9850± 0.0121	0.9354± 0.0217	0.7438± 0.0283	0.7822± 0.0254	67.1	33.5	148.8

used to evaluate the detection performance of our method. The mAP is the average accuracy summation of all the categories divided by all the categories. The AP and mAP for a particular category i are calculated as follows:

$$AP_i = \int_0^1 P(R) dR \quad (15)$$

$$mAP = \frac{1}{N} \sum_{i=1}^N AP_i \quad (16)$$

where $P(R)$ is the mapping relationship between P and R and N is the total number of categories.

3) *Frames Per Second*: FPS indicates the inference speed of the model, measured by the number of image frames processed per second.

4) *Params*: Params denotes the parameters of the object detection model.

5) *GFLOPs*: GFLOPs represent the computational complexity of the model, measured in billions of floating-point operations.

C. Comparison With SOTA Methods

To demonstrate the effectiveness of our method on ACPCs, we compared it with the current SOTA feature similarity-based methods, including Faster RCNN [51], FCOS [52], RetinaNet [53], and Swin Transformer [37]. The experimental results are shown in Table I. To fully ensure the validity of the results, the values in the table represent the mean and standard deviation from fivefold cross-validation. All methods were trained and tested in the same environment, and the results indicate that our method outperforms the others. Specifically, in the missing assembly detection task, we achieved the best results in mAP50, mAP75, mAP50:95, and mAR50:95 (mean average recall at different thresholds). Fig. 8 shows the training loss curves of the proposed method and other methods. According to the comparison results, our method achieved performance comparable to the best training curve method, indicating better generalization on the ACPC test set. We believe that when the randomness of the missing position is strong, the features obtained by the SWA in this article exhibit higher discrimination. This is because the feature embedding vector of the missing position enhances the distinction from the background through affinity, thereby improving the detection and localization of defects. Table I further compares computational efficiency. Although Swin Transformer achieves higher FPS, its computational cost is significantly higher. Our method maintains optimal accuracy while reducing Params

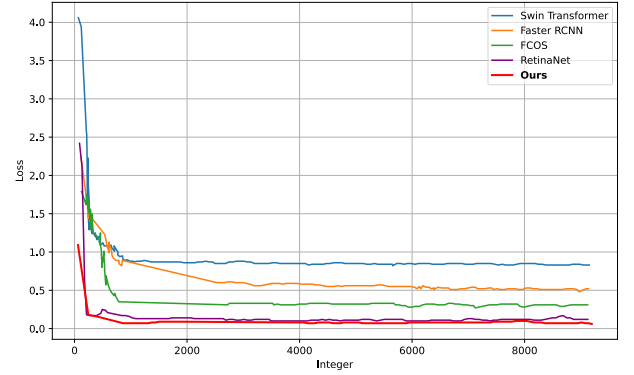


Fig. 8. Training loss curve comparison.

and GFLOPs by 15.8% and 44.6%, enhancing efficiency and practicality.

To visually illustrate the detection efficacy of the model, input test images were randomly chosen as depicted in Fig. 9(a), with the visual outcomes contrasted with those of other methods presented in Fig. 9(b)–(f). Enlargements of the missing position and its adjacent pixels are positioned in the upper left corner of each image. It is evident that our method yields a higher single-image prediction probability and a finer delineation of the detection box location compared to other methods. This is attributed to SWA enhancing the pixel connectivity via affinity, thereby ensuring that pixels in the missing area secure a superior feature representation, which leads to more precise positioning of the detection frame and a discrimination probability that approaches 1.

To further highlight the network's capability to precisely identify areas of missing installation, the test images mentioned above were selected, and the feature maps resulting from feature fusion were visualized, as shown in Fig. 10. In the resulting visualization feature maps, areas that tend toward red indicate a higher attention level by the network, suggesting that these areas are more likely to be detected as missing parts areas. The visualization analysis results clearly demonstrate that the perception ability of the network model proposed in this study significantly surpasses that of other comparative methods. In contrast, the latter methods exhibit excessive attention to normal areas and weaker recognition capability for missing parts areas, indicating these network models' inadequate performance in distinguishing between missing parts areas and normal areas. Furthermore, this phenomenon is attributed to the effectiveness of the SWA module within our network model. The SWA module enhances the correlation among pixels within missing parts areas, making the feature representation of these areas more pronounced and thus more

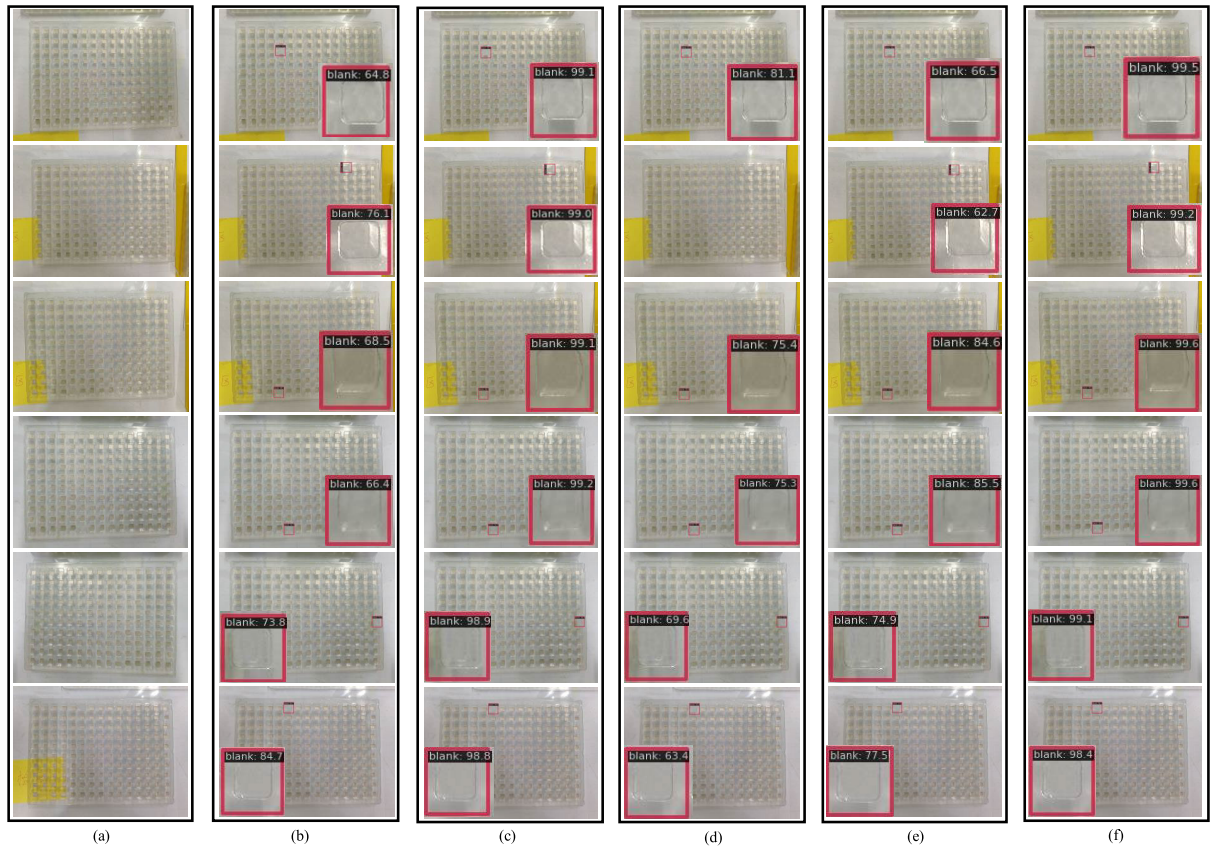


Fig. 9. Visualization and detection results. (a) Original images. (b) Swin Transformer. (c) Faster RCNN. (d) FCOS. (e) Retinanet. (f) Ours.

easily recognized and located by the network. Consequently, it can be concluded that the network model proposed in this study demonstrates higher accuracy and precision in locating missing parts areas on chips.

D. Ablation Experiment

1) *Different Components Proposed*: To rigorously validate the effectiveness of the proposed method, ablation experiments were conducted to compare the contributions of each component. As a control, the CRTA was replaced with a randomly replicated and pasted alternative. The experimental results, as shown in Table II, indicate that the CRTA provided an exceptionally robust data foundation. Building on this, the SWA further enhanced the model's representational capacity through affinity, while the HFP improved the discriminative power between the proactive layer and the supporting layer.

2) *Results for Different Input Images*: To evaluate the impact of different input image resolutions on detection results, images of various resolutions were fed into the network. The results, as presented in Table III. At a resolution of 224, all metrics demonstrated relatively low performance, particularly with map75 and map50:95 values of 0.544 and 0.507, respectively. However, as the input size increased, performance progressively improved, reaching its peak at a resolution of 1300×800 . At this resolution, map50, map50:95, and mar50:95 achieved the highest values of 0.985, 0.743, and 0.782, respectively, indicating superior detection accuracy and robustness. Additionally, although the input sizes of 896×896 and 1792×1792 showed similar performance in

TABLE II
ABLATION RESULTS OF DIFFERENT COMPONENTS

RTA	SWA	HFP	map50	map50:95	mar50:95
			0.941 ± 0.014	0.678 ± 0.022	0.698 ± 0.013
✓			0.954 ± 0.024	0.689 ± 0.014	0.743 ± 0.013
✓	✓		0.964 ± 0.021	0.707 ± 0.018	0.757 ± 0.034
✓		✓	0.965 ± 0.024	0.722 ± 0.022	0.770 ± 0.027
✓	✓	✓	0.985 ± 0.012	0.743 ± 0.028	0.782 ± 0.025

TABLE III
RESULTS FOR DIFFERENT INPUT IMAGES

Input size	map50	map50:95	mar50:95
224×224	0.857 ± 0.024	0.507 ± 0.023	0.566 ± 0.023
448×448	0.959 ± 0.015	0.663 ± 0.026	0.738 ± 0.025
896×896	0.941 ± 0.023	0.708 ± 0.028	0.772 ± 0.031
1792×1792	0.955 ± 0.024	0.699 ± 0.022	0.770 ± 0.021
1300×800	0.985 ± 0.012	0.743 ± 0.028	0.782 ± 0.025

some metrics, the overall performance at 1300×800 was the most outstanding, aligning well with the COCO dataset. On the one hand, this resolution is closer to the actual data distribution, helping to eliminate extraneous redundant information. On the other hand, this size fully leverages the multiscale characteristics of the network structure.

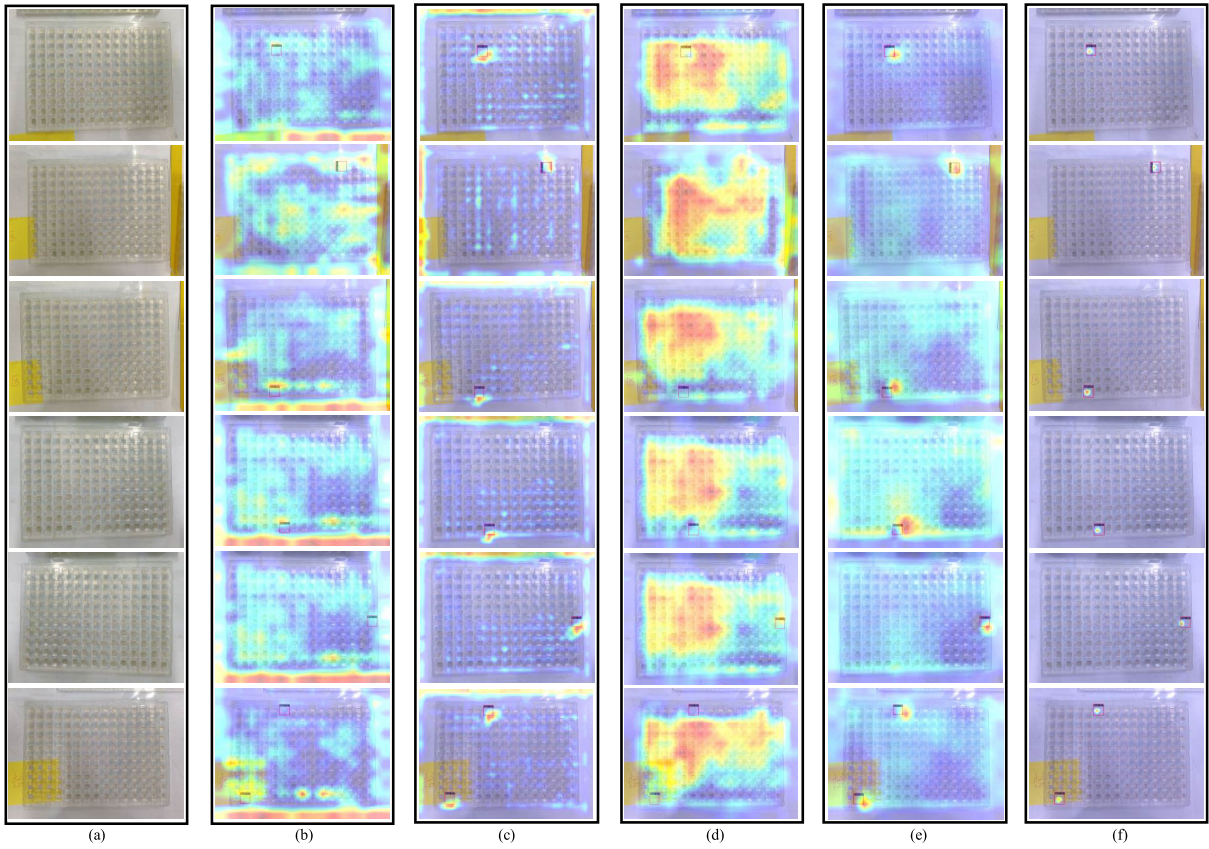


Fig. 10. Visualization effects of feature maps. (a) Original images. (b) Swin Transformer. (c) Faster RCNN. (d) FCOS. (e) Retinanet. (f) Ours.

E. Robustness Experiment

In the actual assembly process, due to changes in the site environment or aging of equipment after long-term operation, noise interference may occur when acquiring ACPC images. To cope with new noises that may be generated in actual production, it is necessary to verify the robustness of ours. The original test set was randomly introduced into four groups of different types of noise, and the strength of each group of noise was compared with each other. The specific test results are shown in Table IV. Among them, the strong noise group includes Gaussian noise with a standard deviation of 10, salt-and-pepper noise with a ratio of 0.007, Poisson noise with an average incidence rate of 25, and speckle noise with a parameter of 0.05. The weak noise group includes Gaussian noise with a standard deviation of 5 and a ratio of 0.005, salt-and-pepper noise, Poisson noise with an average incidence rate of 2, and speckle noise with a parameter of 0.01.

The results show that the different noises introduced in the test set will cause a drop in detection accuracy of the proposed method, but it is still within the acceptable range. Under the most extreme perturbation of strong salt-and-pepper noise, the one with the most change only dropped by 0.124. The proposed method has good environmental adaptability and can cope with various disturbances that randomly appear in the actual assembly inspection process. Strong noise visual effects are shown in Fig. 11. The images in the test set are randomly selected, added with the above strong noise, and sent to the

network for detection, and the detection effect of the model under the noise is intuitively displayed.

To verify the robustness of different models, the test set was added with two sets of salt-and-pepper noise with different intensities and then sent to different models for comparison. It can be seen from Table IV that salt-and-pepper noise has the greatest influence on the accuracy of the model compared with other noise. Therefore, this experiment is representative. The results are shown in Table V. For single-stage detection methods such as FCOS and RetinaNet, the quality of feature extraction has a great influence on the final results, so it is difficult to cope with inclement noise. The Swin Transformer and Faster RCNN, respectively, enhance the feature extraction and candidate box generation, so the declining accuracy is slightly better than the previous two methods. Under the same extreme conditions, the accuracy of our model decreases the least. The correlation between pixels is enhanced by the proposed SWA. Even if the noise is added, the output discriminative features can be output stably.

F. Generalization Experiments

To further evaluate the applicability and detection efficacy of the network model proposed in this study across various detection objects, the NPSS dataset was introduced as a new object for conducting generalization performance experiments. The NPSS dataset [35] is specifically designed for identifying defects in nickel-plated punched steel strips, comprising 1500 images of nickel-plated punched steel strip

TABLE IV
ANTI-INTERFERENCE EXPERIMENT

Noise type	mAP	mAP(75)	mAP Decline ratio
None	0.7380	0.99980	/
Gauss(strong)	0.6580	0.7480	0.08
Gauss(weak)	0.7200	0.9130	0.018
Salt-pepper(strong)	0.6140	0.6810	0.124
Salt-pepper(weak)	0.6590	0.7540	0.079
Poisson(strong)	0.7290	0.9998	0.009
Poisson(weak)	0.7320	0.9998	0.006
Speckle(strong)	0.6700	0.7740	0.068
Speckle(weak)	0.7260	0.9998	0.012

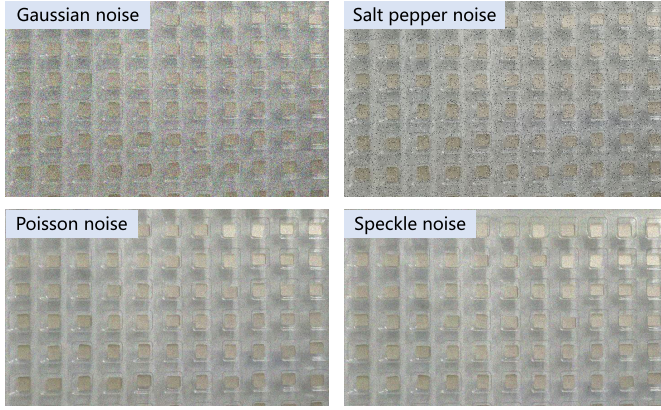


Fig. 11. Strong noise visual effect.

TABLE V
ROBUSTNESS COMPARISON EXPERIMENT

Model	type	mAP	mAP75	mAP decline ratio
Swin Transformer [37]	strong	0.3690	0.2190	0.27
	weak	0.3180	0.3140	0.258
Faster RCNN [51]	strong	0.2990	0.3570	0.382
	weak	0.5630	0.6580	0.118
FCOS [52]	strong	0.0000	0.0000	0.571
	weak	0.0460	0.0500	0.525
RetinaNet [53]	strong	0.0780	0.0700	0.593
	weak	0.2090	0.1750	0.462
Ours	strong	0.6140	0.6810	0.124
	weak	0.6590	0.7540	0.079

defects collected on-site with industrial cameras. This dataset is categorized into five defect types: blind hole (BH), connecting hole (CH), rusty spot (RU), scratch (SC), and water spot (WS), with each category containing 300 images. Due to the use of image stitching technology, these images vary in size.

The experimental comparison results on the NPSS dataset are presented in Table VI, the network model proposed in this study demonstrates exceptional performance across all five categories of nickel-plated punched steel strip defect detection tasks, particularly ranking first in four of the defect detection categories. In terms of the mean mAP metric, the method proposed in this study significantly outperforms other comparative methods. These results validate the proposed network's

TABLE VI
COMPARISON RESULTS WITH VARIOUS FRAMEWORKS
FOR THE NPSS DATASET (%)

MODELS	BH	CH	RU	SC	WS	mAP
YOLOX-m [54]	95.48	99.96	71.92	90.33	72.89	86.12
YOLOX-tiny [54]	94.58	99.51	69.84	88.54	71.88	84.87
YOLOv5-m [55]	95.7	99.92	69.18	95.08	73.55	86.68
EfficientDet-d0 [56]	99.38	99.68	53.01	83.41	69.98	81.09
EfficientDet-d2 [56]	99.69	99.12	65.32	91.5	75.15	86.15
CenterNet [57]	99.84	99.52	61.79	91.33	72.27	84.95
Faster-RCNN [51]	98.84	99.29	57.76	78.3	64.24	79.69
LEDNet [58]	99.96	99.87	73.59	87.69	75.46	87.31
Ours	99.99	99.99	65.87	96.58	75.82	87.65
	(0.021)	(0.032)	(0.041)	(0.018)	(0.015)	(0.034)

The experimental results of this method are the mean of 5-fold cross validation, and the standard deviation of the corresponding indicator is in brackets in the last row.

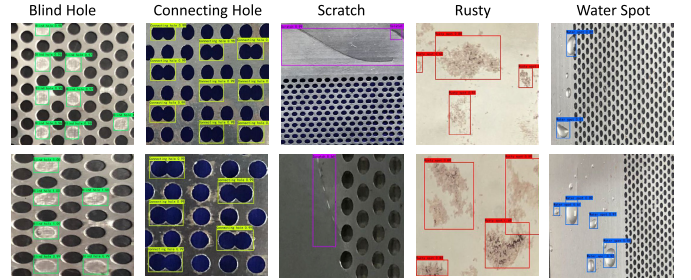


Fig. 12. Visual results of the NPSS dataset using the proposed method.

TABLE VII
RESULTS OF THE PROPOSED MODULE ON THE MVTEC DATASET (%)

Methods	I-AUROC	P-AUROC	PRO
DRaEM [41]	98.0	97.3	—
DRaEM [41]+SSPCAB [42]	98.9	97.2	—
DRaEM [41]+SSMCTB [43]	98.7	97.2	—
PatchCore [44]	99.1	98.1	93.5
SimpleNet [49]	99.6	98.1	91.2
Prompt AD [50]	96.6	96.5	—
Ours	99.1	98.0	92.9

excellent generalization capability, affirming its suitability for a variety of detection scenarios. To intuitively demonstrate the generalization ability of this method, the visual results of the NPSS dataset under this method are shown in Fig. 12. It can be seen that this article has good detection performance in categories such as CH and BH, which have similar structures to the ACPCs dataset proposed in this article. Although the positioning is slightly flawed for irregular categories such as rusty, it can still meet the needs of actual industrial production, further fully verifying the generalization of this article.

The scarcity of defect data and the reliance on supervised learning on labeled defect samples impose significant limitations on supervised methods for industrial defect detection. As industrial defect inspection tasks become increasingly diverse, exploring self-supervised learning-based defect detection has become particularly crucial. Therefore, this paper further investigates the feasibility and adaptability of the core module SWA within an anomaly detection framework.

Given that SimpleNet is the latest SOTA model in the field of anomaly detection and its network architecture facilitates seamless integration of external modules, we select SimpleNet as the anomaly detection baseline and incorporate the SWA module into its structure. Specifically, SWA is inserted after the backbone feature extractor and before the feature adaptor, aiming to enhance feature discriminability. Meanwhile, SimpleNet's loss mechanism remains unchanged to ensure training stability and compatibility.

Comparative experiments conducted on the public dataset MVTec, as shown in Table VII, demonstrate that while the SWA-enhanced SimpleNet does not surpass the original model in overall detection accuracy, it achieves a higher PRO score, validating the effectiveness of the SWA module in anomaly localization tasks and further confirming its potential for unsupervised defect detection.

V. CONCLUSION

In this study, the ACPC dataset was initially constructed. To address the issue of sample imbalance, the data were effectively expanded using the fast region transformation method. Then, a missing assembly detection algorithm based on local window affinity calculation is proposed to calculate the similarity relationship between pixels, strengthen the internal relationship between pixels, highlight the pixel difference between the missing area and the normal area, and use multi-scale feature fusion to enhance the defect feature and improve the discrimination of small object area features. Ultimately, this approach demonstrated significant advantages over other methods and fulfilled industrial production requirements. This research provides a reference for the application of vision technology in intelligent industrial production lines.

However, certain limitations exist and should be addressed in future work. The method relies on a grid-based structural prior, which may limit its applicability to more complex and unstructured industrial assembly scenarios. Additionally, the incorporation of the SWA model and a dual-branch structure introduces additional computational overhead, which may affect real-time performance in high-speed industrial production lines. Future research should focus on enhancing adaptability, optimizing computational efficiency, and broadening applicability to diverse industrial defect detection tasks.

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